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STAT 5010
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Analyzing the U.S. Opioid Crisis

I. Introduction & Background

A. Motivation & Research Questions

We are interested in investigating which economic and environmental factors are most significantly contributing to drug-related deaths in the United States, especially in the context of the opioid crisis. We will examine economic, environmental, and overdose trends for each of the 50 states from 2000 - 2016. Most Americans have been or know somebody who has been personally impacted by the opioid epidemic. Understanding the drivers of overdose and possible mitigation strategies can help public health advocates direct research and funding appropriately, and provide more context for policymakers. This research topic is important because substance abuse and overdose death are alarmingly prevalent in the United States[1].

B. Background

Drug overdose is the leading cause of death for Americans under 50 years old. In 2024, only 1 in 5 people who suffer from drug abuse received treatment. The opioid crisis in the United States has also been exacerbated by an explosion of Fentanyl in 2013[2]. If we could help predict which communities are at risk and why, it could help distinguish where substance abuse treatment resources need to go and identify which factors that lead to substance abuse. This research benefits treatment providers, schools and community organizations, and public health agencies.

C. Data Sources

We integrated three separate sources of data to answer our research question. One dataset comes from the Centers for Disease Control (CDC) and National Center for Health Statistics

[1] Evan D. Gumas, "U.S. Overdose Deaths Remain Higher Than in Other Countries — Trend-Tracking and Harm-Reduction Policies Could Help," To the Point (blog), Commonwealth Fund, Jan. 9, 2025. https://doi.org/10.26099/ppdk-qy10
[2] U.S. Department of Health and Human Services. 2026. Social Determinants of Health. https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health

(NCHS)[3]. The dataset provides solid historical information (2000-2016) on drug deaths in the United States broken down by State, Age Group, Race (White, Hispanic, Black), and Sex (Male, Female). Unfortunately, we were not able to use the demographic information due to aggregation peculiarities; however the data enables us to see trends in drug mortality during the early 21st century and provides the response variables: Death Rate per 100k Population, and Number of Deaths. Another source is University of Kentucky's Center for Poverty Research (UKCPR) National Welfare Data[4]. This contains important socioeconomic factors such as poverty rate, state minimum wage, gross state product (GSP), unemployment rate, etc. The final source is DEA archival reports on retail drug transactions[5]. These reports provided the quantities of various drugs purchased by each state, used as a proxy for drug supply because we do not have insight into the illicit market supply. Combined, this data is a rich set of predictors and responses spanning 17 years. It is ideal for identifying the strongest economic and environmental factors relating to overdose deaths.

D. Approach

We utilized different methods of regression to identify a model that fits the data best, tells us which features predict drug death, and shed light on what kind of individuals are affected most. We will use many statistical methods including multiple linear regression, negative binomial regression, comparing models with the ANOVA test, and fixed-effects models.

II. Methods & Results

A. Method Assumptions

1. Multiple Linear Regression

MLR assumptions include (i) linearity between features and response, (ii) homoskedasticity, (iii) normality of residuals, (iv) no major multicollinearity, (v) independence of errors, and (vi) a continuous variable as response. Our data satisfies (vi) because the response variable is death rate per 100k population, and

[3] National Center for Health Statistics. 2018. NCHS - Drug Poisoning Mortality by State: United States. Centers for Disease Control and Prevention. https://data.cdc.gov/NationalCenter-for-Health-Statistics/NCHS-Drug-Poisoning-Mortality-by-State-United-Stat/xbxb-epbu/about_data Accessed: 2026-04-20

[4] University of Kentucky Center for Poverty Research. 2024. UKCPR National Welfare Data, 1980–2023. University of Kentucky. https://ukcpr.uky.edu/resources Accessed: 2026-04-20.

[5] U.S. Drug Enforcement Administration. 2026. ARCOS Retail Drug Summary Reports: 2000–2005 Archives. Diversion Control Division, U.S. Department of Justice. https://www.deadiversion.usdoj.gov/arcos/retail_drug_summary/archive/archives-report.html Accessed: 2026-04-20.

does not seriously violate (ii) based on Residuals vs. Fitted plots.. However, we failed to satisfy the other assumptions based on visual analysis of diagnostic plots (e.g. Fitted vs. Observed Plots, QQ Plots, Added Variable Plots) and formal statistical tests (e.g. Durbin-Watson Test, Shapiro-Wilk Test. We abandoned MLR methods and moved on to Negative Binomial Regression as a result.

2. Negative Binomial Regression

Negative Binomial Regression assumptions include (i) response is a count variable, (ii) overdispersion, (iii) linearity in the log-link, (iv) independence of individual observations, and (v) no omitted variable bias. Our data satisfies all assumptions with the exception of (iv), independence of individual observations (Durbin-Watson statistic of 0.34). This is due to temporal autocorrelation: observations for Colorado in 2008 are not independent from observations for Colorado in 2007. However, this issue was addressed through clustered standard errors which confirmed that our most important predictors remained significant while accounting for autocorrelation. Our response in this case is the number of deaths (count variable), and visual inspection of partial residual plots confirms the log-linear assumption holds for primary predictors[6]. We confirmed overdispersion with the DHARMa dispersion test and by validating that the mean of the response variable is significantly lower than the variance. Finally, we ensured proper specification of predictors by determining the best model based on various criteria (AICc, BIC, HQIC).

3. Fixed Effects Model

Lastly, we did fixed-effects models, beginning with a one-way fixed effects model. This model specifically demeaned by state therefore assumptions were that the unique, permanent traits of each state were biasing our results. Other key assumptions include no perfect collinearity, linearity in parameters, strict exogeneity, homoscedasticity, and no serial autocorrelation. The data does not follow the assumptions of serial autocorrelation and homoscedasticity. This is because the data involves entries over 17 years thus those assumptions are inherently violated. Then we moved on to two-way fixed effects, controlling for

[6] See Appendix Figure 1

both year and state. This assumes that the year contributed to events that impacted every state at once. The assumptions of the one-way model remain valid for the two-way After this we shifted the drug variables back by one year creating a lagged fixed-effects model. This is testing the timing of cause and effect, assuming that the effects that happened last year explain the changes we see this year.

B. Results

Negative Binomial Regression proved to be a powerful tool in identifying the most important factors leading to drug overdoses, and in predicting future deaths. To derive the optimal model, we fit a full model along with models optimized for minimizing AICc, BIC, and HQIC respectively. Crucially, each model included an offset of the log of population to handle exposure.

Criterion	**Predictors**	**df**	**AICc**	**BIC**	**HQIC**
Full Model	gsp_per_cap, log_fent, log_hydro, log_oxy, medicaid_rate, min_wage, poverty_rate, snap_rate, unempl_rate	11	10997.07	11048.95	11012.93
Min. AICc	gsp_per_cap, log_fent, log_hydro, log_oxy, min_wage, poverty_rate, snap_rate	9	10994.64	11037.13	11006.97
Min. HQIC	gsp_per_cap, log_hydro, log_oxy, min_wage, snap_rate	7	10996	11029.09	11004.78
Min. BIC	**gsp_per_cap, log_hydro, log_oxy, min_wage, snap_rate**	**7**	**10996**	**11029.09**	**11004.78**

We ultimately decided on the model chosen by both HQIC and BIC ("optimal model") because it has fewer predictors than the other models while maintaining essentially the same AICc score. See table below for summary results of the optimal model fit on the full dataset.

Predictor	**Estimate (β)**	**p-value**	**Importance**

SNAP Rate	0.1552	< 0.001	Primary Driver (High)
Oxycodone (log)	0.1484	< 0.001	Primary Driver (High)
Minimum Wage	0.0833	< 0.001	Secondary Driver
GSP per Capita	0.0474	0.0053	Moderate Driver
Hydrocodone (log)	-0.0548	0.0049	Inverse Relationship

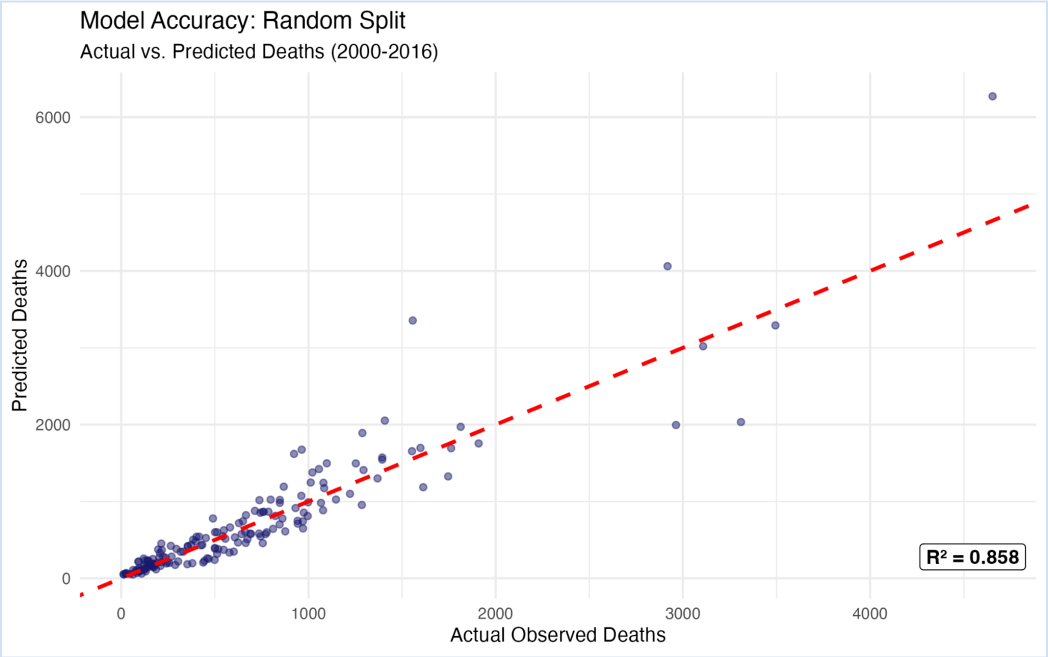
The coefficient estimates and associated significance help answer our primary research question: what are the most significant economic and environmental factors contributing to drug overdose mortality? Through every iteration of model building and testing, SNAP Rate (percentage of people receiving benefits through the Supplemental Nutrition Assistance Program) was consistently the strongest driver of mortality (economic)[7]. Similarly, supply of Oxycodone was consistently the second highest driver of mortality (environmental). Of the other economic predictors, Minimum Wage is the only significant one while GSP per Capita has an insignificant impact on drug deaths. Interestingly, Hydrocodone supply has an inverse relationship with drug deaths. This may reflect a substitution effect where it was phased out in favor of more potent alternatives. Additionally, in 2014 the DEA reclassified Hydrocodone from a Schedule III drug to a Schedule II drug, which has much stricter prescription policies. Oxycodone is also a Schedule II drug but its popularity outweighed any effects of the stricter policies. Ultimately, our results suggest that the crisis is shaped as much by socioeconomic instability as it is by the availability of specific opioids.

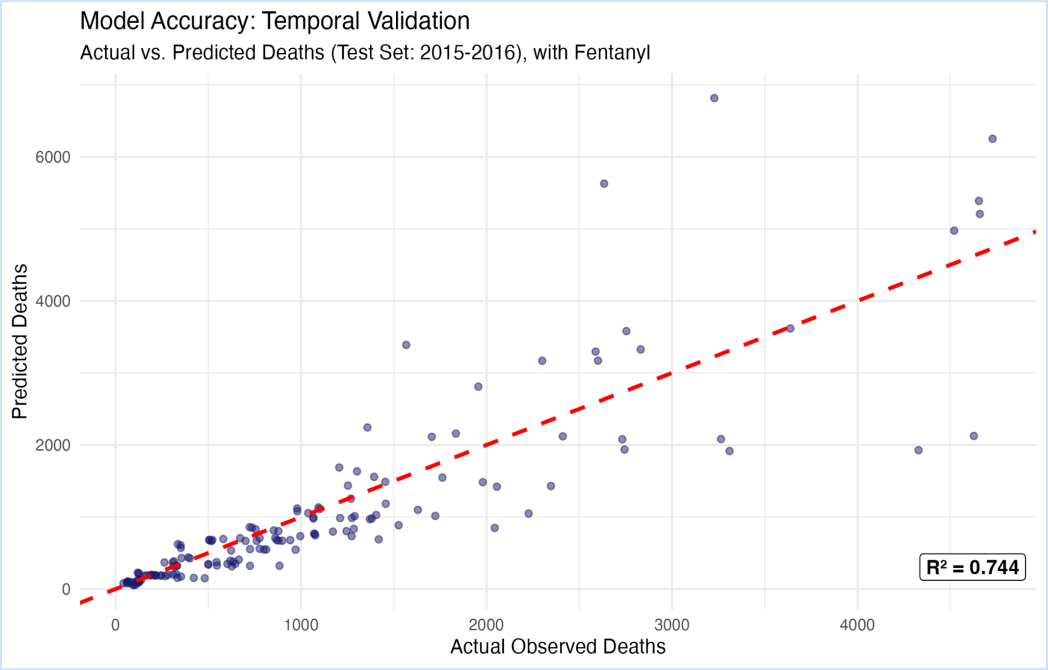
Next, we split our data into training and testing sets and evaluated the model's predictive power. First, we used a random 80/20 split of the entire dataset. Second, we split temporally with training covering 2000-2013 and testing covering 2014-2016. For the temporal split we tested both the optimal model and the optimal model with the addition of Fentanyl supply as a predictor. The reasoning behind adding Fentanyl for the temporal split is that Fentanyl's first big influx to the U.S. was in 2013; we assume that it will be a significant predictor in the years to follow. See table below for results.

[7] See Appendix Figure 2

Split Type	RMSE	MAE	MAPE (%)	Test R^2
Random	299.13	165.25	32.60	0.858
Temporal	662.85	351.84	30.74	0.728
Temporal (+Fentanyl)	633.62	344.65	30.29	0.744

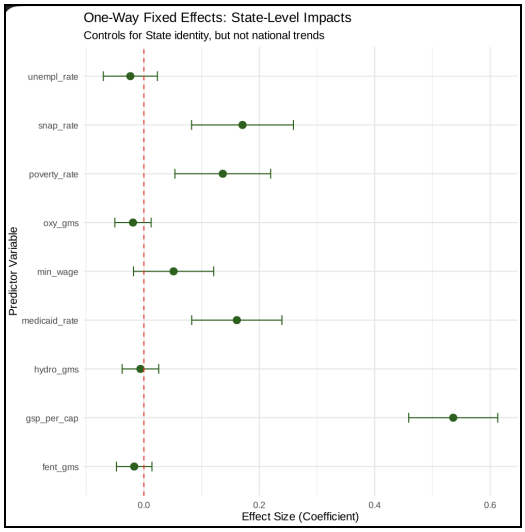
The Random Split results (Test $R^2 = 0.858$) confirm the model's structural components (specifically SNAP enrollment and prescription supply) are highly stable predictors across the 2000-2016 period. To test the model's actual predictive utility, we used the Temporal Split. While the base model maintained a strong fit (Test $R^2 = 0.728$), accuracy was further improved ($R^2 = 0.744$, RMSE decrease of approx. 4.5%) by adding the log of Fentanyl supply as a predictor. This highlights that while economic insecurity is a constant driver of the opioid crisis, the transition toward synthetic opioids post-2013 necessitates adaptive modeling to forecast the future.



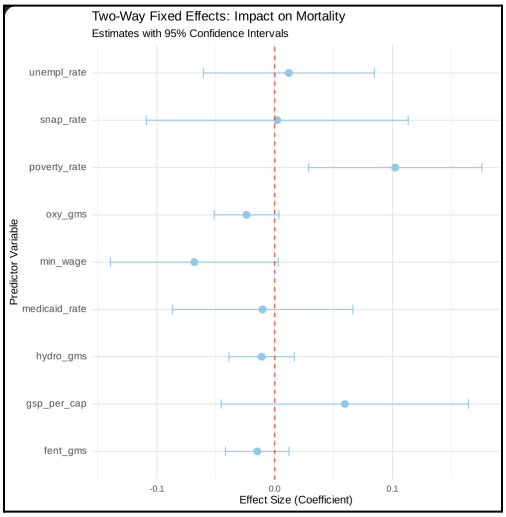


The above charts demonstrate the closer fit of the model when using a Random Split vs. a Temporal Split. Both models struggle to account for high outliers in the number of observed deaths.

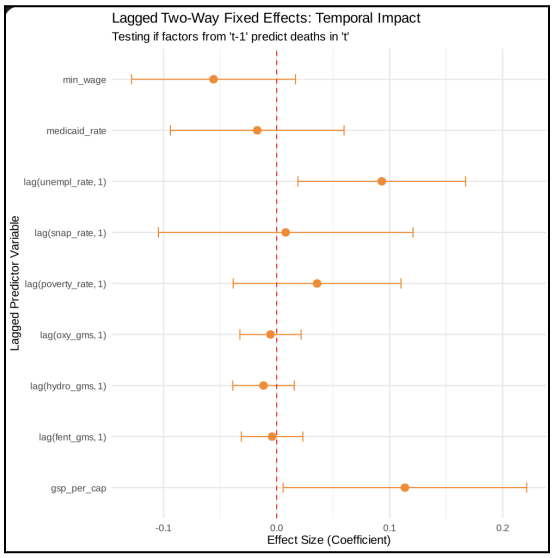
The fixed effects model analysis consisted of a progression of models to isolate the drivers of the opioid-related mortality. The initial one-way fixed effects model which utilized entity demeaning to control for state characteristics, suggested that many socioeconomic factors were strongly correlated with the log-transformed death rate. Specifically, the poverty rate (0.137, $p < 0.01$), the snap rate (0.171, $p < 0.001$), the Medicaid rate (0.161, $p < 0.001$), and the GSP per person (0.526, $p < 0.001$) all showed statistical significance. Additionally we have a R^2 value of 0.662, but this is because over the 17 years of this dataset the Opioid crisis continued to become more prevalent while GSP and Medicaid continued to expand. Therefore, this model sees positive trends and assumes this to be causation when that is not the case.



In contrast, the more rigorous two-way fixed effects model that controlled for national trends, resulted in the GSP, Medicaid, and Snap variables to no longer be statistically significant. The poverty rate remained significant (0.102, $p < 0.01$). This means that once you account for the fact that deaths were rising everywhere each year, the state level supplies do not map to localized death spikes, except for the spike in poverty accurately tracking localized mortality. The R^2 for the two-way model is very low with 0.021, meaning the variables explain about 2.1% of the remaining variability. Interestingly the Adjusted R^2 is -0.072, this is because we are being mathematically punished for adding variables. This does not mean the model failed but that it is penalized and is not forcing over-fitting.



The lagged model was used to shift variables back by one year to see if past conditions further predict current deaths. In this model poverty is no longer significant, however the unemployment rate (0.092, p = 0.014) and GSP per person (0.113, p = 0.0393) were significant. This shows that there is a temporary delay in how macroeconomic shocks hit a population. In specific an increase in unemployment is a warning sign for a possible spike in overdose deaths. The R^2 value for this model is 0.018, meaning our model accounted for almost 2% of the variability present. The Adjusted R^2 is -0.08, which can be explained by the same explanation for the two-way model because of punishment for adding more variables.



III. Conclusions

A. Research Question & Overall Interpretation

Our BIC-optimized model explains 86% of the variance in out-of-sample data and satisfies all formal GLM assumptions for dispersion and mean-variance relationships. Even after penalizing the model for state-level clustering, the SNAP enrollment rate emerges as the most robust predictor of opioid mortality, closely followed by Oxycodone supply.

Our research sought to identify the structural drivers of localized drug-related mortality to better predict at-risk communities. The application of the fixed-effects models led us to find that the opioid crisis is more linked to structural economic distress rather than to the volume of medical prescriptions for these drugs. Once national longitudinal trends were accounted for the public assistant programs and supply of opioids did not significantly predict localized mortality

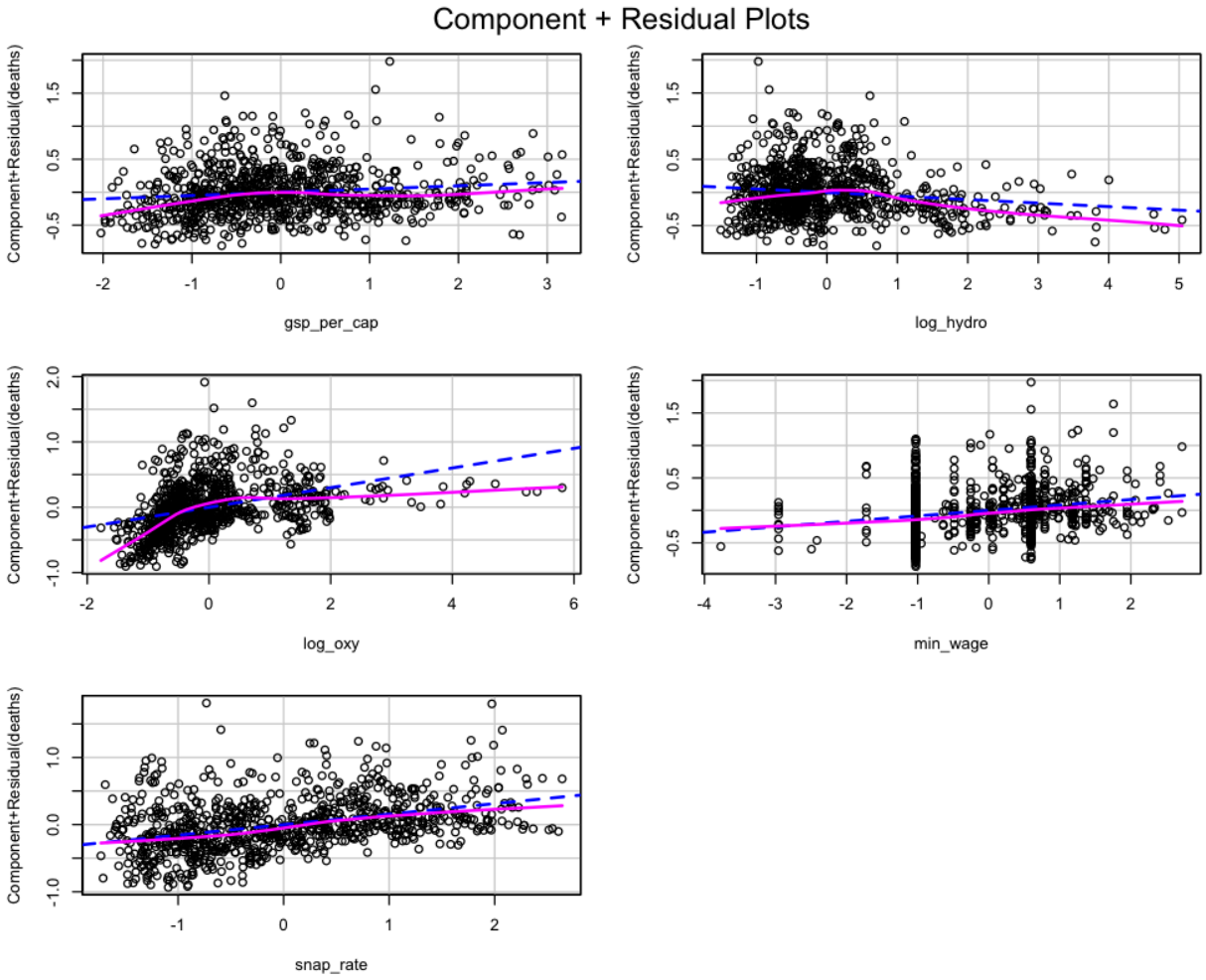
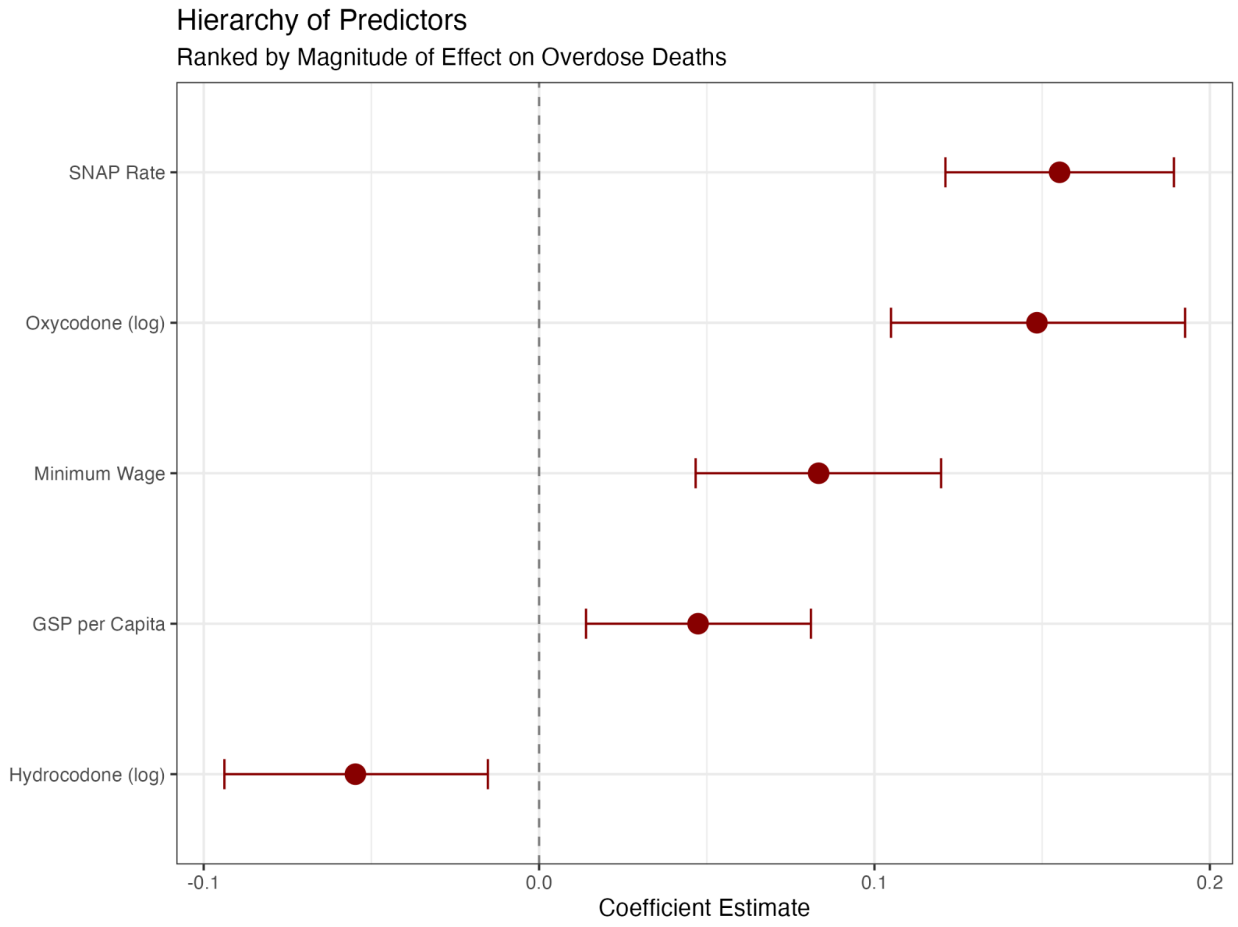
spikes. However, the state-level poverty remained significant, being the immediate driver of overdose deaths. Furthermore, the lagged model showed that while a spike in unemployment does not cause immediate fatalities, it acts as a warning sign to predict deaths for the following year. Ultimately this research gives policymakers concrete evidence and a predictive framework to mitigate the opioid epidemic.

B. Future Recommendations

Based on our findings, we recommend that public health agencies use macroeconomic indicators to create an early warning system to predict drug overdose related deaths. Additionally, future research should examine demographic variables such as race and gender to proactively protect vulnerable populations. It is very important that research continues as the opioid crisis is an ongoing problem that needs more attention.

IV. Appendix

A. Figure 1: Negative Binomial Component + Residual Plots



B. Figure 2: Negative Binomial Coefficient Confidence Intervals